Simple Linear Regression

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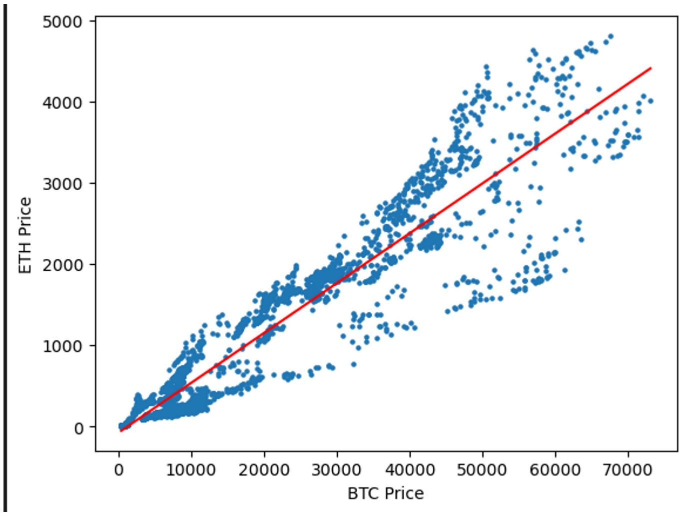


Given a random variable Y and a covariate X

<u>Regression</u> is the process of deducing $E[Y]$ given X.

So we want a function $r(x) = E[Y \mid X = x]$

In <u>linear</u> <u>regression</u> we look at functions of the form

$r(X) = \beta_0 + \beta_1 X$.

<u>Def</u> The simple linear regression model is given by

$Y_i = \beta_0 + \beta_1 X + \varepsilon_i$. Where $E[\varepsilon_i] = 0$ $var[\varepsilon_i] = \sigma^2$ for all ε_i.

we estimate β_0 and β_1 by $\hat{\beta}_0$ and $\hat{\beta}_1$.

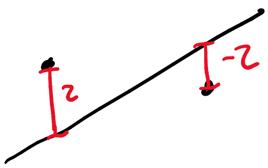
Given that $\hat{Y}_i = \hat{r}(X_i)$ is our predicted value, our residuals are $\hat{\varepsilon}_i = Y_i - \hat{Y}_i$.

The <u>least</u> <u>squares</u> <u>estimate</u> chooses $\hat{\beta}_0$ and $\hat{\beta}_1$ to

The least squares estimate chooses ...
minimize $SS_{res} = \sum_{i=1}^{n} \hat{\varepsilon}_i^2$.

Intuitively: we square so that ① + and − errors don't cancel out.

② more drastic errors are penalized heavily.



More formally If ε_i are normal, then LS is MLE.

In practice we can find $\hat{\beta}_0$ and $\hat{\beta}_1$ by gradient descent. But it is interesting to note that we can solve for $\hat{\beta}_1$ and $\hat{\beta}_0$ explicitly by.

$$\hat{\beta}_1 = \frac{Cov[XY]}{var[X]} \qquad \hat{\beta}_0 = \overline{Y}_n - \hat{\beta}_1 \overline{X}_n$$

Goodness of fit can be evaluated using the Coefficient of determination, R^2.

If $SS_{Res} = \sum_{i=1}^{n} (Y_i - \hat{Y}_i)^2$ and $SS_{tot} = \sum_{i=1}^{n} (Y_i - \overline{Y}_n)^2$

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}.$$

If $SS_{res} = 0 \Rightarrow R^2 = 1$

If $SS_{res} = SS_{tot} \Rightarrow R^2 = 0$.

Least squares and MLE

Goal: Show that if errors are normally distributed, then the least squares line is the MLE for $Y|X$.

$, (z - \mu_i)^2$

line is the MLE for $Y|X$.

Recall • If $Z \sim N(\mu_i, \sigma^2)$ then Z has PDF $f(z) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{z-\mu_i}{\sigma}\right)^2}$

- Simple linear regression: $Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$ with $E[\varepsilon_i] = 0$ and $var[\varepsilon_i] = \sigma^2$.
- To get likelihood you plug your data into the P.D.F and multiply the results.

under the assumption of normality for ε_i, $Y_i \sim N(\beta_0 + \beta_1 X_i, \sigma^2)$

Then the likelihood of Y_i given X_i is

$$L(\beta_0, \beta_1, \sigma) = \prod_{i=1}^{n} f_{Y|X}(Y_i | X_i) \propto \sigma^{-n} \exp\left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^{n} \left(Y_i - (\beta_0 + \beta_1 X_i)\right)^2 \right.$$

Then the log-liklihood is given by

minimize

$$\ell(\beta_0, \beta_1, \sigma) \propto -n \log(\sigma) - \frac{1}{2\sigma^2} \sum_{i=1}^{n} \left(Y_i - (\beta_0 + \beta_1 X_i)\right)^2.$$

$$-n \log(\sigma) - \boxed{\frac{1}{2\sigma^2} \sum_{i=1}^{n} (Y_i - \hat{Y}_i)^2 \quad \text{RSS}}.$$

So maximizing liklihood is minimizing RSS.